

Mini Study Guide 10: The Two-Stage DDM and PVGO

BKM Ch. 18

FIN 4000 Investments · Final exam prep · Sep 28, 2026 · Jim Northey

Learning objectives

A stock's price has two parts: the value of the assets it already has (E_1 / k) and the present value of its growth opportunities (PVGO). Growth adds value only when reinvested earnings earn more than the required return ($ROE > k$).

After this guide you should be able to:

1. Value a stock with the constant-growth DDM and compute sustainable growth ($g = ROE \times b$).
2. Split a stock's price into no-growth value and PVGO.
3. Explain why plowing back earnings can raise, lower, or leave price unchanged.
4. Value a stock with a two-stage DDM, discounting the terminal value correctly.
5. Connect the P/E ratio to growth opportunities.

Core concept

Constant growth and sustainable growth

$$V_0 = \frac{D_1}{k - g} \quad g = ROE \times b \quad D_1 = E_1(1 - b)$$

Where:

- V_0 = intrinsic value of the stock today
- D_1 = dividend expected next year
- k = required rate of return (market capitalization rate)
- g = constant dividend growth rate
- ROE = return on equity
- b = plowback (retention) ratio; $1 - b$ = payout ratio
- E_1 = earnings per share expected next year

b is the plowback (retention) ratio and $1 - b$ is the payout ratio. The formula needs $k > g$. Required return k often comes from the CAPM: $k = r_f + \beta[E(r_m) - r_f]$.

PVGO

$$P_0 = \underbrace{\frac{E_1}{k}}_{\text{no-growth value}} + \text{PVGO} \Rightarrow \text{PVGO} = P_0 - \frac{E_1}{k}$$

Where:

- P_0 = current stock price
- E_1 = earnings per share expected next year
- k = required rate of return
- E_1 / k = no-growth value (the price if all earnings were paid out forever)
- PVGO = present value of growth opportunities

ROE vs. k	Effect of plowing back more earnings	PVGO
ROE > k	Price rises: each reinvested dollar earns more than investors require	Positive
ROE = k	Price unchanged: growth appears, but it adds no value	Zero
ROE < k	Price falls: reinvestment destroys value	Negative

Growth is not the same as growth opportunities. A firm can grow earnings by reinvesting at ROE < k and still lower its stock price.

P/E and growth

$$\frac{P_0}{E_1} = \frac{1}{k} \left(1 + \frac{\text{PVGO}}{E_1/k} \right) = \frac{1-b}{k-g}$$

Where:

- P_0 / E_1 = price-earnings ratio on next year's earnings
- k = required rate of return
- PVGO = present value of growth opportunities
- E_1 = earnings per share expected next year
- b = plowback ratio
- g = growth rate

A high P/E signals that the market expects large growth opportunities relative to current earnings.

The two-stage DDM

Use it when a firm grows fast for T years, then settles into constant growth g_2 forever.

$$V_0 = \sum_{t=1}^T \frac{D_t}{(1+k)^t} + \frac{P_T}{(1+k)^T}, \quad P_T = \frac{D_{T+1}}{k-g_2}$$

Where:

- V_0 = intrinsic value today
- D_t = dividend in year t
- T = number of years of high growth
- P_t = terminal (horizon) price at the end of year T
- D_{t+1} = first dividend of the constant-growth stage
- g_2 = long-run constant growth rate
- k = required rate of return

Steps:

1. Grow the dividend year by year at g_1 through year T .
2. Compute $D_{t+1} = D_t \times (1 + g_2)$.
3. Compute the terminal price $P_t = D_{t+1} / (k - g_2)$. This is a value **at time T** .
4. Discount each dividend and P_t back to today. P_t is discounted T periods, not $T + 1$.

Worked examples

Example 1: PVGO when ROE is above, equal to, and below k (Algorithmic)

A firm expects $E_1 = \$4.00$, retains half its earnings ($b = 0.5$, so $D_1 = \$2.00$), and investors require $k = 10\%$. No-growth value = $E_1 / k = 4 / 0.10 = \$40$.

ROE	$g = \text{ROE} \times b$	$P_0 = D_1 / (k - g)$	$\text{PVGO} = P_0 - \$40$	P/E_1
14%	7%	$2 / 0.03 = \$66.67$	$+\$26.67$	16.7
10%	5%	$2 / 0.05 = \$40.00$	$\$0.00$	10.0
8%	4%	$2 / 0.06 = \$33.33$	$-\$6.67$	8.3

All three firms grow, but only the one earning more than k on new investment has positive PVGO. At $\text{ROE} = 8\%$, the firm would be worth more (\$40) if it paid out every dollar.

Example 2: Two-stage DDM (Algorithmic)

$D_0 = \$1.50$. Dividends grow 20% a year for 3 years, then 5% forever. $k = 11\%$.

Year	Dividend	Terminal value	PV at 11%
1	$1.50 \times 1.20 = \$1.80$		\$1.62
2	$1.80 \times 1.20 = \$2.16$		\$1.75
3	$2.16 \times 1.20 = \$2.592$	$P_3 = 2.592 \times 1.05 / (0.11 - 0.05) = \45.36	$(2.592 + 45.36) / 1.11^3 = \35.06
Value			\$38.44

BA II Plus: CF 2nd CLR WORK · 0 ENTER ↓ · 1.8 ENTER ↓ ↓ · 2.16 ENTER ↓ ↓ · 47.952 ENTER · NPV · 11 ENTER ↓ CPT → 38.44

Put P_3 in the same cash flow as D_3 (year 3), not in year 4.

Example 3: Required return from the CAPM (Algorithmic)

$r_f = 4\%$, $\beta = 1.2$, market risk premium = 6%. $D_1 = \$2.24$ and $g = 5\%$.

1. $k = 4\% + 1.2 \times 6\% = \mathbf{11.2\%}$.
2. $V_0 = 2.24 / (0.112 - 0.05) = \mathbf{\$36.13}$.

If the stock trades at \$32, it is underpriced by this model: its implied return $D_1 / P_0 + g = 7.0\% + 5\% = 12.0\%$ exceeds the 11.2% required.

Common pitfalls

- **Using D_0 in the numerator.** The DDM uses next year's dividend: $D_1 = D_0(1 + g)$.
- **Discounting the terminal value one period too many.** P_t is already a time- T value; discount it T periods.
- **Computing P_t with D_t .** It uses $D_{t+1} = D_t(1 + g_2)$.
- **Equating growth with value creation.** Only $\text{ROE} > k$ creates positive PVGO.
- **Using E_0 for PVGO.** No-growth value is E_1 / k .

Key terms and definitions

- **Intrinsic value:** the present value of a firm's expected future cash payoffs, discounted at the required return.
- **Dividend discount model (DDM):** values a stock as the present value of all expected future dividends.
- **Constant-growth (Gordon) model:** $V_0 = D_1 / (k - g)$, for dividends growing at a constant rate forever.
- **Plowback (retention) ratio and payout ratio:** the share of earnings reinvested (b) vs. paid out as dividends ($1 - b$).
- **Sustainable growth rate, $g = ROE \times b$:** the growth rate a firm can support from reinvested earnings alone.
- **Present value of growth opportunities (PVGO):** the part of a stock's price that comes from future investments earning more than k .
- **No-growth value (E_1 / k):** a firm's value if it paid out all earnings and never grew.
- **Multistage (two-stage) DDM:** a DDM with a high-growth period followed by constant growth.
- **Terminal (horizon) value:** the value at the end of the forecast period of all dividends after that date.
- **Price-earnings ratio:** stock price $\div$ earnings per share; reflects growth opportunities and risk.
- **Market capitalization rate (k):** the market's consensus required return on a stock, often estimated with the CAPM.

CFA-style practice questions

Q1 [Algorithmic]. A stock's next dividend is \$3.00, its required return is 11%, and its dividend growth rate is 5%. Its intrinsic value is *closest* to:

- A. \$27.27
- B. \$50.00
- C. \$52.50

Q2 [Algorithmic]. A firm has a 12% ROE and retains 40% of its earnings. Its sustainable growth rate is:

- A. 4.8%
- B. 7.2%
- C. 12.0%

Q3 [Algorithmic]. A stock sells for \$80. Next year's expected EPS is \$5.00, and the required return is 10%. Its PVGO is *closest* to:

- A. \$30
- B. \$50
- C. \$75

Q4 [Conceptual]. A firm's ROE is 8% and its required return is 10%. If it increases its plowback ratio, its stock price will *most likely*:

- A. rise, because growth increases.
- B. stay the same.
- C. fall, because reinvestment earns less than investors require.

Q5 [Algorithmic]. $D_0 = \$2.00$. Dividends grow 15% for 2 years, then 4% forever. $k = 10\%$. The stock's value is *closest* to:

- A. \$34.67
- B. \$38.72
- C. \$42.17

Q6 [Conceptual]. In a two-stage DDM whose high-growth phase lasts T years, the terminal value P_t should be:

- A. discounted back T years.
- B. discounted back $T + 1$ years.
- C. added to D_{t+1} and discounted back $T + 1$ years.

Q7 [Algorithmic]. A firm has a plowback ratio of 0.40, a required return of 10%, and a growth rate of 6%. Its P/E ratio (on next year's earnings) is:

- A. 10
- B. 15
- C. 25

Q8 [Conceptual]. A stock with a much higher P/E than its industry *most likely* has:

- A. a higher required return than its peers.
- B. a large PVGO relative to the value of its existing assets.
- C. a lower ROE than its required return.

Q9 [Algorithmic]. $r_f = 4\%$, $\beta = 1.2$, and the market risk premium is 6%. A stock's $D_1 = \$2.24$ and $g = 5\%$. Its value is *closest* to:

- A. \$36.13
- B. \$37.33
- C. \$44.80

Answer key

Q	Answer	Explanation
1	B	$3.00 / (0.11 - 0.05) = \$50.00$. C wrongly grows the given D_1 again; A ignores growth.
2	A	$g = 0.12 \times 0.40 = 4.8\%$. B uses the payout ratio (0.60).
3	A	No-growth value = $5 / 0.10 = \$50$. PVGO = $80 - 50 = \$30$.
4	C	With $ROE < k$, each reinvested dollar is worth less than a dollar to shareholders, so PVGO falls.
5	C	$D_1 = 2.30$, $D_2 = 2.645$, $P_2 = 2.645 \times 1.04 / 0.06 = 45.85$. $V = 2.30 / 1.10 + (2.645 + 45.85) / 1.10^2 = \42.17 . B discounts P_2 one year too many; A ignores the high-growth stage.
6	A	P_t is the value at time T of all dividends from $T + 1$ on, so it is discounted T years.
7	B	$P/E = (1 - b) / (k - g) = 0.60 / 0.04 = 15$. C forgets the payout ratio.

Q	Answer	Explanation
8	B	A high P/E reflects large growth opportunities relative to earnings. A would lower the P/E; C implies negative PVGO.
9	A	$k = 4\% + 1.2 \times 6\% = 11.2\%$. $V = 2.24 / (0.112 - 0.05) = \36.13 . C uses $k = 10\%$ (β of 1).